TG-300 Telemetry Gateway

Installation and configuration guide for firmware 3.2

1 Overview

The TG-300 collects readings from field sensors over Modbus RTU and forwards them to a central broker over MQTT. This guide covers mounting, first start-up, network configuration and the most common faults. It is written for installers who are familiar with basic Linux commands.

2 Requirements

2.1 Hardware

Item	Minimum	Recommended
Power supply	12 V DC, 1 A	24 V DC, 2 A with surge protection
Enclosure	IP54	IP66 with sun shield
Network	Ethernet 10/100	Ethernet plus LTE backup modem
Serial line	RS-485, two-wire	RS-485, shielded twisted pair, terminated

2.2 Software

The gateway runs a hardened Linux image that is installed at the factory. You only need a laptop with an SSH client and a web browser to complete the set-up.

3 Installation

1. Mount the gateway on a DIN rail inside the enclosure, away from frequency converters.
2. Connect the RS-485 line to terminals A and B.
   - Use a 120 ohm terminating resistor at the last device.
   - Keep the cable shield grounded at one end only.
3. Connect the Ethernet cable and apply power; the status LED blinks green while the system starts.
4. Log in over SSH with the installer account printed on the label inside the lid.

After the first login, install the latest package set and enable the collector service:

Listing 1. Update and enable the collector

```
sudo tg-update --channel stable
sudo systemctl enable --now tg-collector.service
systemctl status tg-collector.service --no-pager
```

> WARNING: Do not disconnect power while an update is running. An interrupted update leaves the gateway in recovery mode and requires a site visit.

4 Configuration

All settings are kept in a single YAML file at /etc/tg300/gateway.yaml. Changes take effect when the collector service is restarted.

4.1 Example configuration

Listing 2. Minimal gateway.yaml

```
site:
  id: PUMP-STATION-07
  timezone: Europe/Lisbon
modbus:
  port: /dev/ttyS1
  baudrate: 19200
  parity: even
  poll_interval_s: 15
mqtt:
  host: broker.example.net
  port: 8883
  tls: true
  topic_prefix: utilities/water/pumps
```

4.2 Parameters

Parameter	Type	Default	Description
`site.id`	string	—	Unique name of the site, used in every topic.
`modbus.baudrate`	integer	9600	Serial speed; must match all field devices.
`modbus.poll_interval_s`	integer	30	Seconds between two complete polling cycles.
`mqtt.tls`	boolean	true	Encrypts the connection to the broker.
`mqtt.topic_prefix`	string	tg300	First part of every published topic.
`buffer.max_mb`	integer	256	Disk space used to store readings while offline.

5 Troubleshooting

5.1 No readings arrive at the broker

Check that the collector service is running and that the broker host name resolves from the gateway. If the LED blinks orange, the gateway is buffering readings because the uplink is down.

5.2 Modbus timeouts

Timeouts usually come from a wrong baud rate or from a missing terminating resistor. Reduce the polling interval only after the line has been checked with a handheld tester.

For further help, contact the support desk and quote the serial number printed on the side of the unit.